

ANASIEZE IKENNA – CLOUD ENGINEER

Project: Refactoring Apps with AWS Managed Services

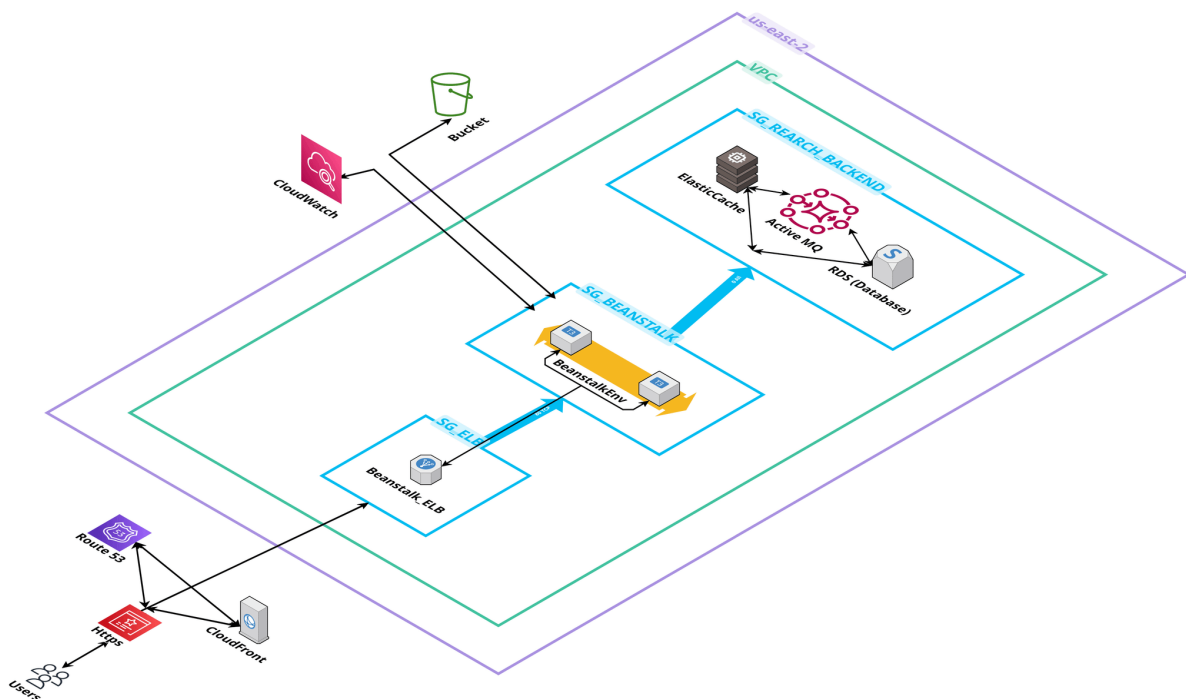
Overview

In this project, I refactored the Profile multi-tier application from a lift-and-shift deployment into a cloud-native architecture using AWS managed services. The goal was to eliminate the operational overhead of managing servers, improve agility, and ensure the application could scale seamlessly for production workloads.

• AWS Managed Services Used

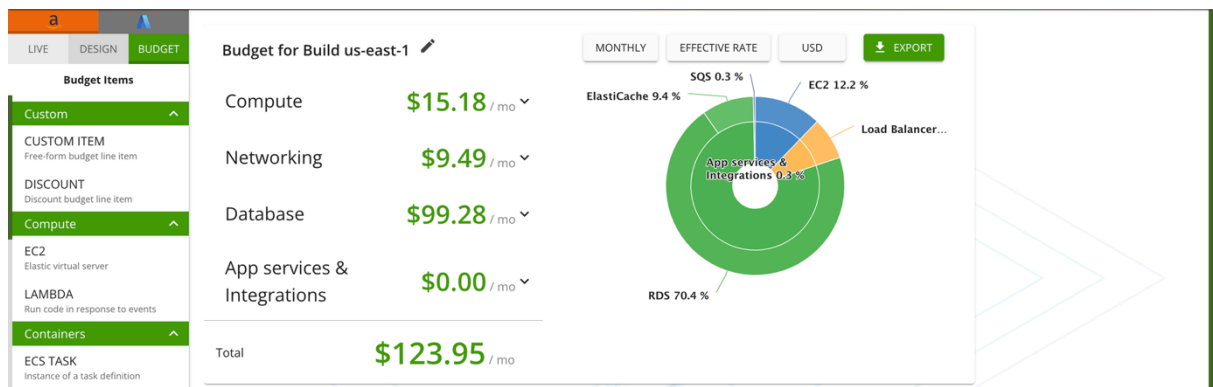
- Elastic Beanstalk → Replaces Tomcat on EC2, handling app deployment, load balancing, and auto scaling.
- Amazon RDS → Replaces MySQL on VM/EC2, providing managed relational database service with backups and scaling.
- Amazon ElastiCache → Replaces Memcache on EC2 for caching.
- Amazon MQ → Replaces RabbitMQ on EC2 for messaging.
- Amazon Route 53 → DNS management for application endpoints.
- Amazon CloudFront → CDN for global content delivery with HTTPS.
- Amazon S3 → Artifact storage and static content delivery.
- Amazon CloudWatch → Monitoring and scaling automation.

Architecture Overview



• Architecture Cost

Running the full Profile stack using AWS Managed Services cost around ~\$123.95/month, showing how AWS Managed Services delivers higher scalability, availability, security and reliability at a higher level.



• Service Modernization

Traditional Service	AWS Managed Service	Benefit
Tomcat on EC2	Elastic Beanstalk	Automated deployment, scaling, patching
MySQL on VM/EC2	Amazon RDS	Backups, failover, scaling built-in
Memcache on EC2	Amazon ElastiCache	Managed caching cluster
RabbitMQ on VM/EC2	Amazon MQ	Fully managed messaging
NFS	Amazon EFS / S3	Elastic storage
DNS	Route 53	High-availability DNS
CDN	CloudFront	Global caching, SSL

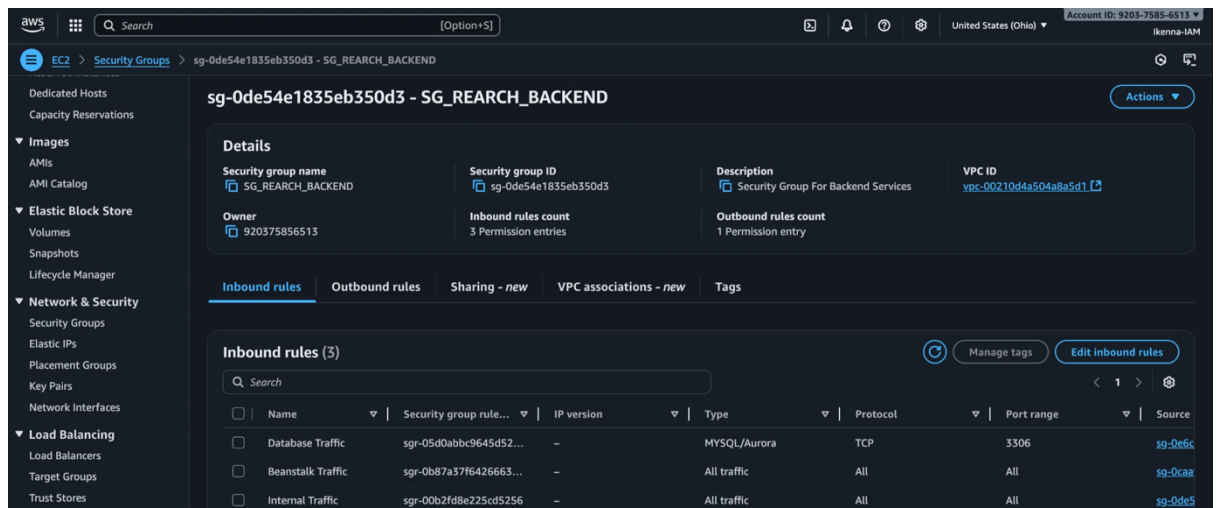
• Flow

1. Users access the app through a Route 53 DNS record.
2. Requests are routed through CloudFront CDN, providing cached content to global users.
3. CloudFront forwards traffic to the Application Load Balancer, part of Elastic Beanstalk.
4. Elastic Beanstalk auto-scales Tomcat EC2 instances running the application.
5. Artifacts are stored in Amazon S3 for deployment.
6. Backend services: RDS (database), ElastiCache (cache), Amazon MQ (messaging).
7. CloudWatch monitors resources and triggers auto scaling events.

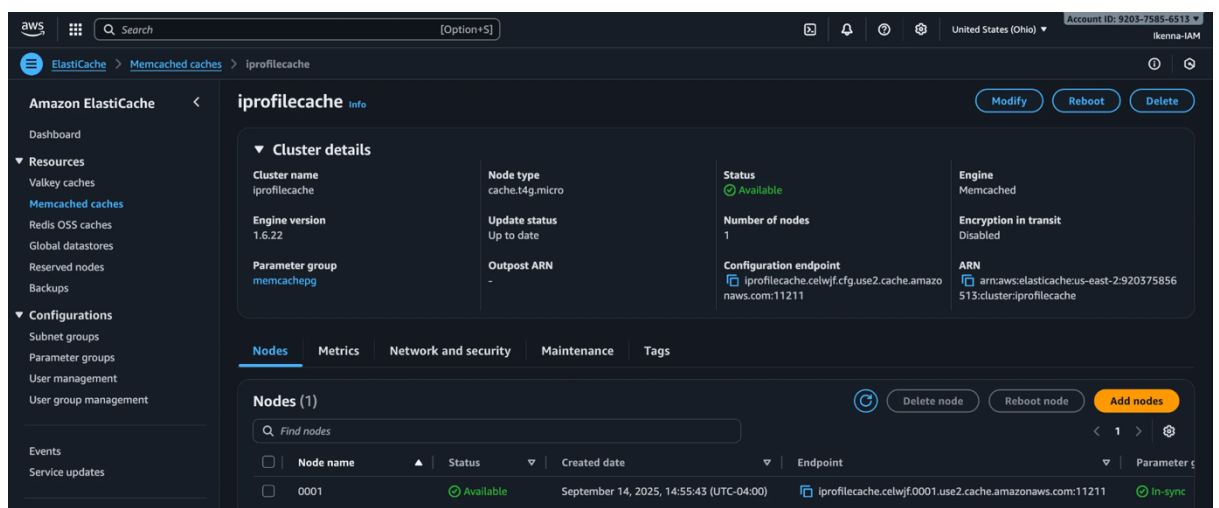
• Why This Project Matters

Traditional lift-and-shift migrations reduce CapEx but still carry operational overhead. By refactoring with AWS managed services, I demonstrated how to cut manual maintenance, improve uptime, and scale globally — turning a legacy stack into a **cloud-native, business-ready application**.

- **Implementation Steps**
 - Re-architected infrastructure from IaaS (EC2-heavy) to PaaS/SaaS managed services.
 - Created an Elastic Beanstalk environment for application servers with built-in scaling and load balancing.
 - Deployed RDS, ElastiCache, and Amazon MQ for database, caching, and messaging layers.
 - Configured CloudFront CDN with SSL certificates for secure HTTPS and global reach.
 - Linked everything with Route 53 DNS and strict security group rules.
 - Automated scaling and health monitoring with CloudWatch alarms and logs.
- **Set Up**
 - Create Security Group For Backend Services



- Create Backend Services (ActiveMQ, RDS, ElastiCache)



Amazon MQ

Brokers

Configurations

ActiveMQ

Details

Specifications

ARN [Info](#)
arn:aws:mq:us-east-2:920375856513:broker:ActiveMQ:b-99b09e69-86d5-456e-9a95-7464e34b7b27

Broker name
ActiveMQ

Broker status
Running

Broker instance type [Info](#)
mq.t3.micro

Deployment mode [Info](#)
Single-instance broker

Broker engine [Info](#)
RabbitMQ

Broker engine version
3.13.7

Creation time (Local)
September 14, 2025 at 15:12 (UTC-4:00)

Configuration

Configuration name
ActiveMQ-configuration

Configuration revision
Revision 1 - Auto-generated default for ActiveMQ-configuration on RabbitMQ 3.13 [Info](#)

CloudWatch Logs
General
Disabled - [Logs](#)

Security and network

VPC [Info](#)
vpc-00210d4a504a8a5d1 [Info](#)

Subnet(s) [Info](#)
subnet-0d139338249142074 [Info](#)

Security group(s) [Info](#)
sg-0de54e1835eb350d3 [Info](#)

Public accessibility [Info](#)
No

Maintenance

Automatic minor version upgrade
Yes

Maintenance window
Wednesday 21:00 - 23:00 UTC

CloudShell

Feedback

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Aurora and RDS

Databases

Query editor

Performance insights

Snapshots

Exports in Amazon S3

Automated backups

Reserved instances

Proxies

iprofilersdatabase

Summary

DB identifier
iprofilersdatabase

Status
Available

Role
Instance

Engine
MySQL Community

Recommendations

CPU
3.12%

Class
db.t3.micro

Current activity
0 Connections

Region & AZ
us-east-2c

Connectivity & security

Monitoring

Logs & events

Configuration

Zero-ETL integrations

Maintenance & backups

Data migrations

Connectivity & security

Endpoint & port

Endpoint
iprofilersdatabase.ccsn8tau9kd6.us-east-2.rds.amazonaws.com

Port
3306

Networking

Availability Zone
us-east-2c

VPC
vpc-00210d4a504a8a5d1

Subnet group
rds_subnet

Subnets
subnet-0d139338249142074
subnet-018ad7981a4943549
subnet-0ccd1410ccb46411c

Security

VPC security groups
SG_REARCH_BACKEND (sg-0de54e1835eb350d3)
Active

Publicly accessible
No

Certificate authority [Info](#)
rds-ca-rsa2048-g1

Certificate authority date
May 21, 2061, 20:04 (UTC-04:00)

DB instance certificate expiration date
September 14, 2026, 14:21 (UTC-04:00)

CloudShell

Feedback

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○ Create Elastic Beanstalk Service

Elastic Beanstalk > **Environments** > **IprofileBeanstalk-env**

Environment overview

Health: Ok

Environment ID: e-ureituuqvx

Domain: iprofile.us-east-2.elasticbeanstalk.com

Application name: Iprofile Beanstalk

Platform

Platform: Tomcat 10 with Corretto 21 running on 64bit Amazon Linux 2023/5.7.4

Running version: Iprofile Beanstalk-version-1.version1.1

Platform state: Supported

Events (19)

Time	Type	Details
September 15, 2025 07:29:46 (UTC-4)	INFO	Environment health has transitioned from Pending to Ok. Initialization completed 17 seconds ago and took 4 minutes.
September 15, 2025 07:29:46 (UTC-4)	INFO	Added instances [i-0768e8f2c9550746d, i-0847b33ff61d5adc2] to your environment.
September 15, 2025 07:29:35 (UTC-4)	INFO	Successfully launched environment: IprofileBeanstalk-env

○ Build & Deploy Artifacts

```
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 4.189 s
[INFO] Finished at: 2025-09-15T12:14:04-04:00
[INFO] -----
apple@MacBookPro iprofile-project %
```

Elastic Beanstalk > **Environments** > **IprofileBeanstalk-env**

Environment overview

Health: Ok

Environment ID: e-ureituuqvx

Domain: iprofile.us-east-2.elasticbeanstalk.com

Application name: Iprofile Beanstalk

Platform

Platform: Tomcat 10 with Corretto 21 running on 64bit Amazon Linux 2023/5.7.4

Running version: Iprofile Beanstalk-version-1.version1.1

Platform state: Supported

Overall health

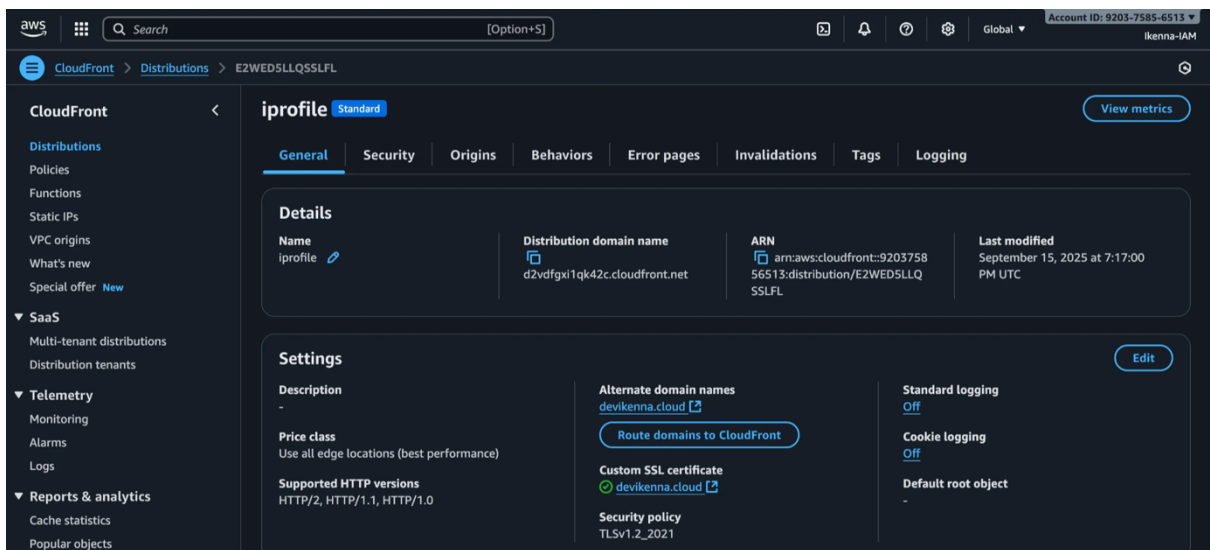
Requests / second	2XX responses	3XX responses	4XX responses	5xx responses
0.4	4	-	-	-

P99 latency(ms)	P90 latency(ms)	P75 latency(ms)	P50 latency(ms)	P10 latency(ms)
7.5	7	6.5	6	4

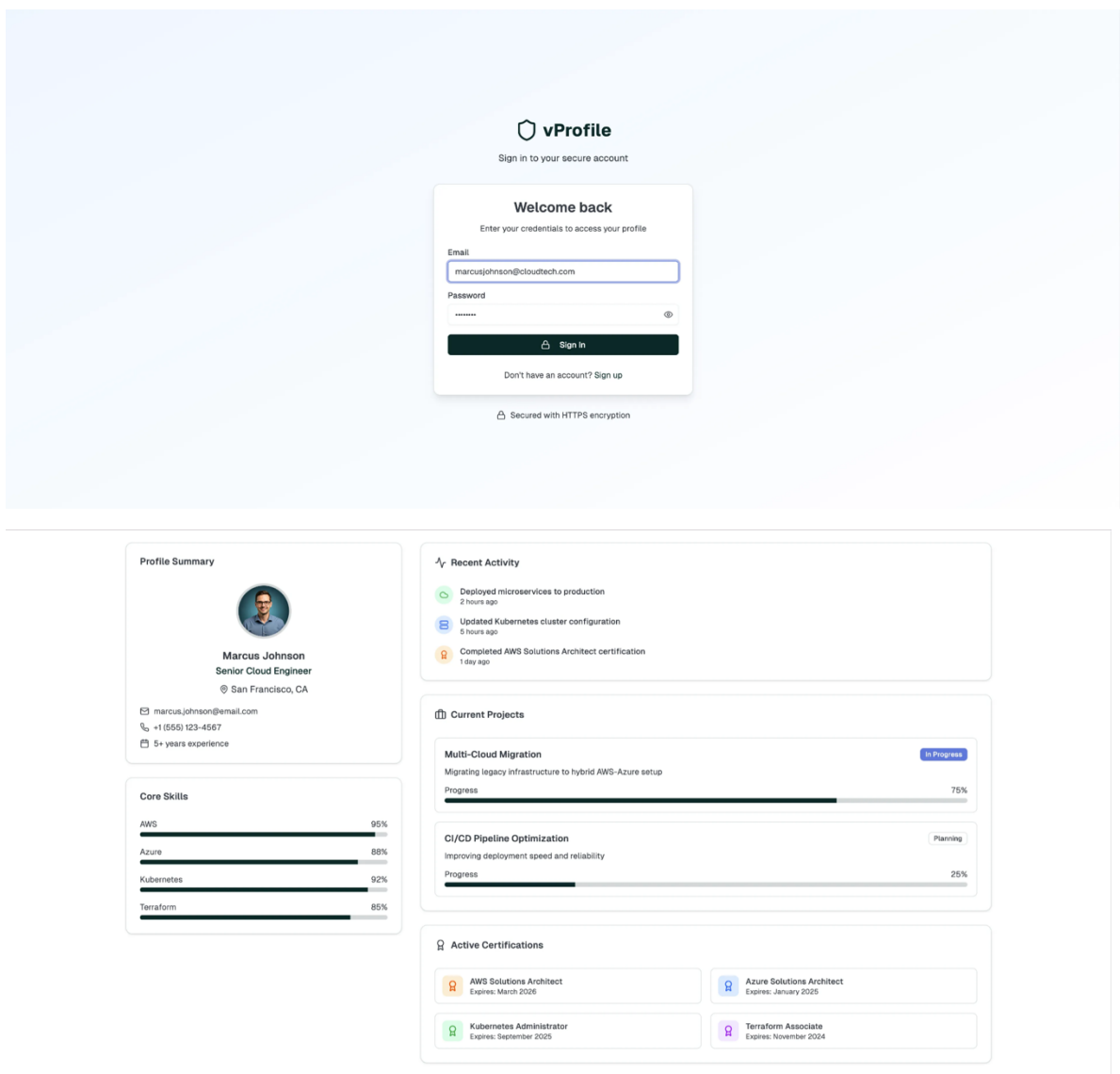
Enhanced instance health (2)

Instance ID	Status	Running time	Deployment ID	Requests/sec	2xx Responses	3xx Responses	4xx Responses	5xx Responses
i-0847b33ff61d5adc2	Ok	5 hours, 34 mi...	2	0.2	2	-	-	-
i-0768e8f2c9550746d	Ok	5 hours, 34 mi...	2	0.2	2	-	-	-

○ Setting up CloudFront



○ Solution Deployed & Running



- **Key Benefits**

- Reduced Ops Overhead by ~60% → Eliminated manual patching, backups, and scaling by moving to managed services.
- Faster Deployments (~70% improvement) → Elastic Beanstalk + S3 enabled app artifact deployment in minutes instead of hours.
- Improved Uptime (99.99% target) → Auto-healing, RDS failover, and managed caching reduced downtime risk significantly.
- Global Reach (~40% lower latency) → CloudFront CDN delivered cached content closer to users worldwide.
- Cost Control (~30% savings) → Pay-as-you-go managed services cut CapEx and reduced ongoing maintenance costs.
- Agility → Freed engineering teams from ops tasks, allowing more focus on feature delivery instead of firefighting.

- **Future Improvements**

- Convert infrastructure to IaC using Terraform or AWS CDK
- Explore container-based deployments using ECS or EKS

This project demonstrates my ability to not just migrate workloads to AWS (Lift & Shift), but to refactor them into managed, cloud-native services — unlocking agility, scalability, and operational efficiency.